

CSE 3407 Notes - Analysis of Algorithms

Michael Lee

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1 Syllabus / How to Class Works

- Prof. Kunal Agrawal
- Need to watch a video lecture *BEFORE* class
- Quizzes every Monday morning based on the video lecture. Unlimited attempts
- No lecture notes :(
- Monday is lecture, Wednesday is recitation (i.e. in-class problems)
- Miss recitations $\Rightarrow$ must make up during OH within 6 days
- HW is 2 problems. HW is graded for completion. Graded out of 1.

2 Greedy Algorithms

2.1 What is a Greedy Algorithm?

- The ‘correct’ greedy algorithm is the solution that finds the solution in the shortest amount of time.
- A ‘greedy’ choice (GC) is a choice you make that is the best choice at the current time. This leads to a smaller subproblem that is the same as the previous problem
- Follow this 3-step proof strategy (basically induction):
 1. **Def.** Greedy Choice Property (GCP) - Some optimal solution is the ‘correct’ greedy choice. Need to prove that some optimal solution will make your greedy choice. This is where you make an ‘Exchange Argument.’
 2. **Def.** Inductive Substructure (IS) - After first choice, you get a smaller instance of the same problem.
 3. **Def.** Optimal Substructure (OS) - IF we solve the smaller instance optimally and combine it with GC, we get an optimal solution overall.

2.2 Ex. Minimum Spanning Tree (MST) Generation

- Suppose we have a connected undirected graph, $G = (V, E)$ where $|V| = n$ and $|E| = m$
- Each edge, $e \in E$, has a unique weight, $w(e) \in \mathbb{R}_{>0}$
- GOAL: Select a set of edges, T , s.t.
 1. The graph is connected
 2. $|T| = \sum_{e \in T} w(e)$ is minimized

Proof. **Claim 1:** T is a tree. Suppose T is not a tree. Thus, there is a cycle in T . However, if T contains a cycle and we remove an edge, e , we get a tree, T' s.t. $|T'| < |T|$.

$\rightarrow \leftarrow$

Claim 2: Find the minimum spanning tree (MST) of G .

Greedy Choice Pick $e \in E$ with endpoints (u, v) s.t. $\min\{w(e)\}$.

Consider some MST, $T^* = (U^*, E^*)$. There are two cases:

1. **Case 1:** $e \in T^*$, so $u, v \in V^*$
2. **Case 2:** $e \notin T^*$. Let $w(z, w)$ have endpoints (z, w) and $w(z, w) > w(u, v)$.

$$\bar{T} = T^* + (u, v) - (z, w) \Rightarrow w(\bar{T}) = w(T^*) - w(z, w) + w(u, v) \leq w(T^*)$$

Hence, $\bar{T}$ is an MST containing (u, v) .

Inductive Substructure: After first choice, we get a smaller instance of the same problem. Just find the edge with the smallest weight s.t. at least one of its endpoints are not in the tree already. In other words,

$$ST(G) = ST(G') + (u, v)$$

Optimal Substructure: This results in a spanning tree because $ST(G')$ is a tree, so $\exists$ a path, p , connected all $a, b \in V(ST(G'))$. There is also a path between all $a \in V(ST(G'))$ and v as the path is simply $P : a \rightarrow u \rightarrow v$. Hence, $ST(G)$ is a tree. $\square$

- The greedy algorithm for MST generation is called **Kruskal's algorithm**, which does the following:
 1. Sort the edges by weight, $w(e) \leftarrow O(m \log(n))$
 2. Picks edges with smallest weight into the tree, T . ($O(1)$)
 3. Removes all edges s.t. adding them creates a cycle.
 4. Repeat
- Total runtime is $O(m \log(n))$